

SUDARSHAN ARYAL

sudarshanaryal99@gmail.com | +91 93812 85688 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Substack](#)

Profile

Undergraduate researcher in Biosciences & Biotechnology with combined wet-lab and computational experience in clinical genomics, NGS pipeline development, and microbial/viral phylogenetics. Builds tools that function, documents rigorously, and does not move on until the underlying biology is understood. Also writes on evolutionary biology and ecology for a general audience. Seeking graduate training in genomic epidemiology and computational biology.

Research Interests

Genomic epidemiology · clinical & microbial bioinformatics · comparative genomics & phylogenetics · NGS pipeline development

Technical Skills

- **Bioinformatics & Pipelines:** fastp, BWA-MEM, SAMtools, MultiQC, Primer3, Tricycler, Autocycler, Polypolish, Bakta, Prokka, BUSCO, Filtlong, Snippy, Gubbins, RAxML, FastTree, IQ-TREE, AutoDock Vina, UCSF Chimera, PyMOL
- **Environment:** Unix/Linux command-line workflows
- **Wet-Lab:** clinical DNA/RNA extraction, multiplex PCR primer design, Illumina library preparation, Qubit / TapeStation QC, Illumina NextSeq X Plus sequencing

Education

B.S. (Hons.) with Research, Biosciences & Biotechnology

2023 – 2027

Sri Sathya Sai Institute of Higher Learning, India

CGPA: **8.7 / 10**

Senior Secondary (Science)

2020 – 2022

St. Xavier's College, Kathmandu, Nepal

GPA: **3.62 / 4.0**

Research Experience

Research Intern, Cancer Genomics & Clinical Bioinformatics

May – June 2026

Karkinos Healthcare Pvt. Ltd., Navi Mumbai, India

Supervisor: Dr. Paras Sehgal, Senior Scientist, Karkinos Healthcare

Academic Supervisor: Dr. B.E. Pradeep, Associate Professor, SSSIHL

- Designed and open-sourced the **HPV Broad-Spectrum Primer Discovery Pipeline** [[GitHub](#)], a multiplex PCR primer design tool applying Alpha-papillomavirus phylogeny stratification (Alpha-5/7/9/10), Primer3-style thermodynamic constraints, 3'-terminal mismatch enforcement, and greedy set-cover optimisation across 224 PaVE L1 reference sequences.
- Produced a validated 4-plex cocktail covering **23 HPV genotypes**, including 11 IARC Group 1 high-risk types, with FFPE-compatible amplicon sizing (120–220 bp).
- Built an end-to-end **NGS HPV L1 genotyping pipeline** (fastp / BWA-MEM / SAMtools / MultiQC) with MAPQ-aware alignment to a curated 224-type HPV reference panel, automated per-sample coverage depth and breadth analysis, and multi-sample genotype-level summary reporting.
- Benchmarked GP5+/GP6+ L1 consensus primers on a clinical Indian cervical cohort ($n = 5$ clinical + 1 positive control); achieved HPV-specific mapping rates of 25–71%, resolved **mixed infections** across 12 genotypes (HPV16, 18, 31, 33, 35, 45, 53, 59, 81, 85, 86, 90) at $\geq 1\%$ read threshold, and confirmed HPV16 positive-control purity at 97.3% with $>87,000\times$ mean coverage depth.
- Executed complete wet-lab NGS workflow independently: clinical DNA extraction (QIAamp DNA Blood Mini Kit), HPV-targeted PCR amplification, DNA and RNA library preparation (TWIST Enzymatic Fragmentation 2.0), Qubit fluorometric and Agilent TapeStation quality control, equal-mass library pooling (250 ng per library), and Illumina NextSeq X Plus 2 $\times$ 150 bp paired-end sequencing.

Projects

Hybrid Genome Assembly of *Brucella anthropi*

April 2026 [\[GitHub\]](#)

ONT long reads + Illumina short reads · Trycycler · Autocycler · Polypolish · Bakta · Prokka · BUSCO · fastp · Filt-long

- Assembled a high-quality *B. anthropi* genome via ONT–Illumina hybrid strategy using Trycycler/Autocycler consensus assembly, achieving high BUSCO completeness after iterative Polypolish error correction; annotated with Bakta and Prokka.

Comparative Genomics & Phylogenetics of *Brucella*

2026 [\[GitHub\]](#)

Snippy · Gubbins · RAxML · FastTree · IQ-TREE

- Performed recombination-aware core SNP calling (Snippy / Gubbins) and inferred maximum-likelihood phylogenies (IQ-TREE / RAxML) across multiple *Brucella* genomes; resolved host-range divergence patterns with >95 % bootstrap support.

Protein–Ligand Docking Study

2024

AutoDock Vina · UCSF Chimera · PyMOL

- Modelled protein-ligand binding poses and ranked interactions by affinity score using AutoDock Vina; visualised key binding residues and surface geometry via PyMOL.

Training & Workshops

- **BioAsia 2026**, Hyderabad — Delegate; sessions spanning biotech innovation, translational genomics, and Asia-Pacific health technology policy.
- **FABA Biotechnology Training** (in association with Thermo Fisher Scientific) — 10-day intensive covering cell culture, aseptic technique, and industry NGS workflows.
- **National Cadet Corps (NCC)** — Training in leadership, field discipline, and team coordination.

Science Communication

Substack — Science & Nature Writing

2024 – Present

- Authored 9 essays at the intersection of evolutionary biology, ecology, and natural philosophy; uses writing as a tool to think carefully about living systems and communicate that thinking to a general audience.

References — Dr. Paras Sehgal (Senior Scientist, Karkinos Healthcare) and Dr. B.E. Pradeep (Associate Professor, SSSIHL) — available on request.